

**Code No: 134AP****JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B.Tech II Year II Semester Examinations, April - 2018****DATABASE MANAGEMENT SYSTEMS****(Common to CSE, IT)****Time: 3 Hours****Max. Marks: 75****Note:** This question paper contains two parts A and B.

Part A is compulsory which carries 25 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b, c as sub questions.

**PART- A****(25 Marks)**

- 1.a) How to represent the strong Entity set and Weak entity set in ER-Model? [2]
- b) Explain about various integrity constraints in relational model. [3]
- c) What are the SQL statements are used to retrieve and modify the database? [2]
- d) Let  $R = (ABC)$  and  $S = (DEF)$  let  $r(R)$  and  $s(S)$  be relations on schema R and S. Give an expression in the Domain relational calculus that is equivalent to each of the following.  
i)  $\sigma_{B=25}(r)$       ii)  $\prod_{A,F}(\sigma_{C=D}(rXs))$  [3]
- e) What is schema refinement? [2]
- f) Define Multi valued dependencies and join dependency. [3]
- g) What is serializability? [2]
- h) Explain Failure with loss of nonvolatile storage. [3]
- i) What is primary and secondary indexing? [2]
- j) What is the difference between indexing and hashing? [3]

**PART-B****(50 Marks)**

2. Give an overview of database architecture. [10]
- OR**
- 3.a) Give an overview of database languages – DDL and DML.
  - b) What are speciality databases? Explain. [5+5]
- 4.a) Explain the fundamental operations in relational algebra with examples.
  - b) What aggregate operators does SQL support? Explain with examples. [5+5]
- OR**
- 5.a) What is trigger? Explain how to implement triggers in SQL?
  - b) Explain the following Operators in SQL with examples:  
i) SOME      ii) IN      iii) EXCEPT      v) UNION. [5+5]

- 6.a) What do you mean by scheme refinement? Explain how it can be accomplished?  
b) What are the problems caused by redundancy and decomposition of relation? [5+5]

**OR**

- 7.a) Compute the closure of the following set of functional dependencies for a relation scheme.  $R(A,B,C,D,E,F,G,H)$ ,  $F=\{ AB \rightarrow C, BD \rightarrow EF, AD \rightarrow G, A \rightarrow H \}$   
List the candidate keys of R.  
b) Explain 4NF, 5NF normal forms with examples. [5+5]

- 8.a) What is transaction? Explain the properties of transaction.  
b) Give an overview of validation based protocol. [5+5]

**OR**

- 9.a) Explain about the Multiple granularity Concurrency Control protocol.  
b) Explain about remote backup system. [5+5]

- 10.a) Give a comparison of various file organizations.  
b) Describe the Insertion and Deletion Operations in B+ trees. [5+5]

**OR**

11. How does *Extendable hashing* use a directory of buckets? How does it handle the insert and delete operations? [10]

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